



Subnetting Revision — From Zero

We'll use this network throughout:

192.168.1.0/24

Imagine this is a company network with:

Tokyo LAN A

Tokyo LAN B

Toronto LAN A

Toronto LAN B

1. First: What is an IP address?

An IPv4 address has **32 bits**, divided into four octets:

192 . 168 . 1 . 0
8 8 8 8 = 32 bits

Each octet can contain:

0 → 255

So:

192.168.1.0

is simply one IPv4 address.

2. What is a network?

A network is a **group of IP addresses that belong to the same logical network.**

For example:

192.168.1.0/24

means:

Network address: 192.168.1.0

Usable hosts: 192.168.1.1 → 192.168.1.254

Broadcast: 192.168.1.255

So this network contains:

256 total addresses

but:

254 usable host addresses

because two addresses have special purposes.

3. Network Address

This was one of your earlier doubts.

Network address = identifies the network itself.

For:

192.168.1.0/24

the network address is:

192.168.1.0

It does **not** identify a computer.

Think:

Network = Apartment building

Network address = Building's identity

Host address = Individual apartment

4. Broadcast Address

The broadcast address is used to communicate with **all hosts in that subnet**.

For:

192.168.1.0/24

broadcast is:

192.168.1.255

For example, if a computer sends a broadcast packet to:

192.168.1.255

all devices in that subnet can receive it.

Important:

Broadcast $\neq$ Default Gateway

5. Default Gateway

The **default gateway** is normally the router's interface IP on that LAN.

Example:

Network: 192.168.1.0/24

Gateway: 192.168.1.1

PC1: 192.168.1.10

PC2: 192.168.1.20

If PC1 wants to communicate with something **outside its own subnet**, it sends the packet to:

192.168.1.1

That is the gateway.

So remember:

Network address → identifies network

Host address → identifies device

Broadcast → all devices in subnet

Gateway → router interface

6. What is a Subnet?

Now comes the important part.

Suppose we have:

192.168.1.0/24

It gives us one large network:

192.168.1.0

↓
0 ————— 255

But imagine we don't want one huge network.

We want:

Tokyo LAN A

Tokyo LAN B

Toronto LAN A

Toronto LAN B

So we **divide the big network into smaller networks**.

Those smaller networks are called **subnets**.

7. What does "borrowing bits" actually mean?

This was your biggest doubt earlier.

/24 means:

24 network bits

8 host bits

The last octet is:

00000000

All 8 bits are currently available for hosts.

Suppose we want **4 subnets**.

We need:

$$2^2 = 4$$

So we take 2 of those 8 host bits and use them to distinguish the subnets.

Before:

NNNNNNNN NNNNNNNN NNNNNNNN HHHHHHHH

After:

NNNNNNNN NNNNNNNN NNNNNNNN NNHHHHHH

That's what **borrowing 2 bits** means.

We didn't physically move anything.

We simply changed the role of those bits.

Now:

Network bits = 26

Host bits = 6

Therefore:

/26

8. Why does **/26** create 4 networks?

The two subnet bits can be:

00

01

10

11

That's:

4 combinations

Therefore:

$2^2 = 4$ subnets

Our original:

192.168.1.0/24

becomes:

192.168.1.0/26

192.168.1.64/26

192.168.1.128/26

192.168.1.192/26

Notice something important:

The subnet mask tells us where the boundaries are.

9. Why does **.64** become another network?

This was another question you asked.

For **/26**:

Host bits = 6

Therefore each subnet has:

$2^6 = 64$ addresses

So the blocks are:

0 — 63

64 — 127

128 — 191

192 — 255

Therefore:

192.168.1.0/26

covers:

192.168.1.0 → 192.168.1.63

Next:

192.168.1.64/26

covers:

192.168.1.64 → 192.168.1.127

So .64 is a new network because the /26 mask creates a **64-address block**.

10. Network, Hosts and Broadcast

Take:

192.168.1.64/26

Its block is:

64 → 127

Therefore:

Network address = 64

First host = 65

Last host = 126

Broadcast = 127

Complete:

192.168.1.64 ← Network

192.168.1.65

192.168.1.66

...

192.168.1.126 ← Last host

192.168.1.127 ← Broadcast

So:

Total addresses = 64
Usable hosts = 62

11. CIDR

CIDR means:

Classless Inter-Domain Routing

Instead of writing:

255.255.255.192

we can write:

/26

They mean the same thing.

Some important ones:

CIDR	Subnet Mask	Total Addresses	Usable Hosts
/24	255.255.255.0	256	254
/25	255.255.255.128	128	126
/26	255.255.255.192	64	62
/27	255.255.255.224	32	30
/28	255.255.255.240	16	14
/29	255.255.255.248	8	6
/30	255.255.255.252	4	2

12. Why /27 has 30 hosts

You specifically struggled with this earlier.

/27:

$32 - 27 = 5$ host bits

Therefore:

$2^5 = 32$ total addresses

But:

1 → Network

1 → Broadcast

Therefore:

$32 - 2 = 30$ usable hosts

Example:

192.168.1.64/27

Range:

64 → 95

Therefore:

64 Network

65

66

...

94 Last host

95 Broadcast

13. Subnet Mask

The CIDR prefix corresponds to a subnet mask.

For example:

/27

means:

11111111.11111111.11111111.11100000

Convert each octet:

11111111 = 255

11111111 = 255

11111111 = 255

11100000 = 224

Therefore:

/27 = 255.255.255.224

14. Fixed-Length Subnetting

Now we reach the difference you were asking about.

Suppose:

192.168.1.0/24

and you want 4 equal subnets.

You borrow 2 bits:

/24 → /26

You get:

Subnet 1: 192.168.1.0/26

Subnet 2: 192.168.1.64/26

Subnet 3: 192.168.1.128/26

Subnet 4: 192.168.1.192/26

Every subnet has:

64 total addresses

62 usable hosts

This is **fixed-length subnetting**.

All subnets use the same mask.

15. VLSM

Now imagine:

Tokyo LAN A → 110 hosts

Toronto LAN B → 45 hosts

Toronto LAN A → 29 hosts

Tokyo LAN B → 8 hosts

If you use **/26** everywhere:

62 usable hosts per subnet

Tokyo LAN A needs 110.

✗ Doesn't fit.

And giving every LAN **/25** would waste lots of addresses.

So we use **VLSM**.

VLSM = Variable Length Subnet Mask

Different LANs can have different subnet sizes.

For our example:

LAN	Required	CIDR
Tokyo A	110	/25
Toronto B	45	/26
Toronto A	29	/27
Tokyo B	8	/28

Then allocate:

192.168.1.0/25
192.168.1.128/26
192.168.1.192/27
192.168.1.224/28

Notice how they **don't overlap**.

That's why there is no collision.

16. Point-to-Point Connection

You also mentioned that your topology has a **point-to-point connection** between routers.

A point-to-point link usually needs only **two usable IP addresses**.

For example:

```
Router A ----- Router B
  |                   |
192.168.1.241       192.168.1.242
```

A **/30** subnet gives:

4 total addresses

2 usable hosts

Example:

192.168.1.240/30
192.168.1.240 → Network
192.168.1.241 → Router A
192.168.1.242 → Router B
192.168.1.243 → Broadcast

That's why **/30** is commonly associated with traditional point-to-point IPv4 links.



Your Entire Subnetting Topic in One Picture

Think of your original:

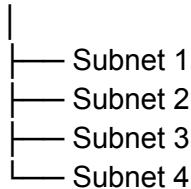
192.168.1.0/24

as a **big apartment building**.

Subnetting:

BIG NETWORK

192.168.1.0/24



CIDR tells us **how big each network is**.

Subnet mask tells us **where the boundary is**.

Network address tells us **which network it is**.

Host addresses identify **individual devices**.

Broadcast address reaches **all devices in that subnet**.

Gateway is the **router interface used to leave the subnet**.

VLSM allows **different-sized subnets**.

Point-to-point links can use a **small subnet such as /30**.



The formulas you should remember

Number of total addresses

$2^{(\text{host bits})}$

Usable hosts

$$2^{(\text{host bits})} - 2$$

Number of subnets when borrowing n bits

$$2^n$$

Host bits

$$32 - \text{CIDR prefix}$$

For example:

$$/27$$

$$\begin{aligned}\text{Host bits} &= 32 - 27 \\ &= 5\end{aligned}$$

$$\begin{aligned}\text{Total addresses} &= 2^5 \\ &= 32\end{aligned}$$

$$\begin{aligned}\text{Usable hosts} &= 32 - 2 \\ &= 30\end{aligned}$$